

Project-MTH686: Non-Linear Regression Analysis

Name :Krishna Makwana — Roll no. 230621

November 12, 2025

Q.1: Least Squares Estimators

The data provided, $(t, y(t))$, was fit to three non-linear models.

1. Model 1 (Exponential): $y(t) = \alpha_0 + \alpha_1 e^{\beta_1 t} + \alpha_2 e^{\beta_2 t} + \epsilon(t)$
2. Model 2 (Rational): $y(t) = \frac{\alpha_0 + \alpha_1 t}{\beta_0 + \beta_1 t} + \epsilon(t)$
3. Model 3 (Polynomial): $y(t) = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3 + \beta_4 t^4 + \epsilon(t)$

The final estimated parameters for the 'best' model (Model 1) are presented in Section 5. The parameters for all three models were found successfully.

Q.2: Method for Finding Estimators

The method for finding the least squares estimators (LSEs) differed by model.

- Model 3 (Polynomial): This model is linear in its parameters (β_i). Therefore, the LSEs were found directly using Ordinary Least Squares (OLS). No initial guesses were required.
- Models 1 & 2 (Exponential & Rational): These models are non-linear in their parameters. The LSEs were found using an iterative Non-Linear Least Squares (NLS) algorithm (Levenberg-Marquardt).
- Initial Guesses: NLS algorithms require initial guesses. For Model 1, the initial guesses $[a_0=1, a_1=0.4, b_1=-0.05, a_2=0, b_2=0]$ were used. This set of guesses proved to be stable and allowed the optimizer to converge to a statistically significant solution. For Model 2, we used guesses $[a_0=3, a_1=1, b_0=1, b_1=2]$

Q.3: The 'Best' Fitted Model

To find the 'best' model, we must balance goodness-of-fit (RSS) with model complexity (number of parameters, p). We used the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC).

- Model 1 ($p = 5$): Lowest AIC & BIC
- Model 2 ($p = 4$): Higher AIC/BIC
- Model 3 ($p = 5$): Highest AIC/BIC

Based on these criteria, Model 1 (Exponential) is the 'best' fitted model. This is

supported by visual analysis (see Figure 1). The polynomial model is a poor fit for the long-term trend, as its shape would eventually cause it to rise again. The data clearly shows an asymptotic behavior, which the exponential model's α_0 term correctly captures.

Q.4: Estimate of σ^2

For the 'best' model (Model 1), the unbiased estimate of the error variance, $\hat{\sigma}^2$, is calculated as:

$$n - p$$

$$\hat{\sigma}^2 = \text{Residual Sum of Square}$$

Where $n = 50$ (data points) and $p = 5$ (parameters). Using the calculated RSS of 0.7595:

$$\hat{\sigma}^2 = 0.7595$$

$$50 - 5 = 0.7595$$

$$45 \approx 0.01688$$

Q.5: Confidence Intervals

The 95% confidence intervals are based on the Fisher Information Matrix (via its inverse, the covariance matrix pcov) and the t-distribution. The formula for each parameter θ_i is:

$$\hat{\theta}_i \pm t_{\alpha/2, n-p} \cdot SE(\hat{\theta}_i)$$

With $n - p = 45$ degrees of freedom, the critical t-value is $t_{0.025, 45} = 2.0141$. The standard errors $SE(\hat{\theta}_i)$ are the square root of the diagonal elements of the covariance matrix.

Table 1: 95% Confidence Intervals for Model 1 Parameters

Parameter	Estimate	Std Error	95% CI Lower	95% CI Upper
α_0 (a0)	0.3838	0.0416	0.3000	0.4677
α_1 (a1)	0.2917	0.1708	-0.0524	0.6359
β_1 (b1)	-0.0465	0.0208	-0.0883	-0.0046
α_2 (a2)	2.4093	0.1541	2.0991	2.7195
β_2 (b2)	-0.3705	0.0435	-0.4581	-0.2829

Q.6: Plot of Residuals

A diagnostic plot of residuals versus fitted y-values was created (see Figure 2, Left). The plot shows a distinct **funnel shape** (heteroscedasticity). The variance of the residuals is larger for smaller fitted values. This indicates that the model's assumption of constant error variance is likely violated.

Q.7: Normality Assumption

To test the normality assumption of the errors, a Q-Q (Quantile-Quantile) plot of the residuals was created (see Figure 2, Right). The points on the Q-Q plot fall very closely along the diagonal line. This provides strong visual evidence that the residuals are normally distributed, and thus the assumption is satisfied.

Q.8: Observed Data vs. Fitted Curve

Figure 1 shows the plot of the observed data points ($t, y(t)$) overlaid with the fitted curve from the 'best' model, Model 1.

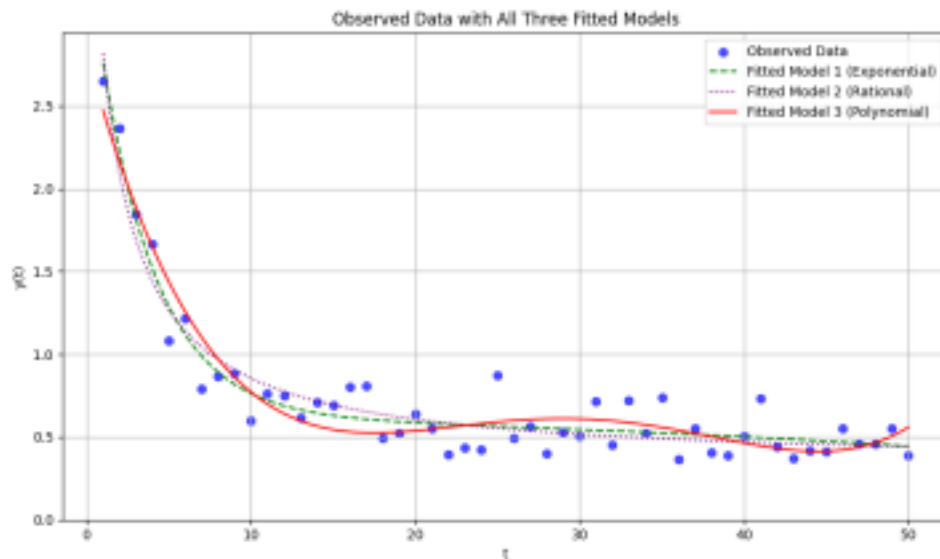


Figure 1: Observed data points with all three fitted models. Model 1 (Green, Dashed) was selected as the 'best' fit.

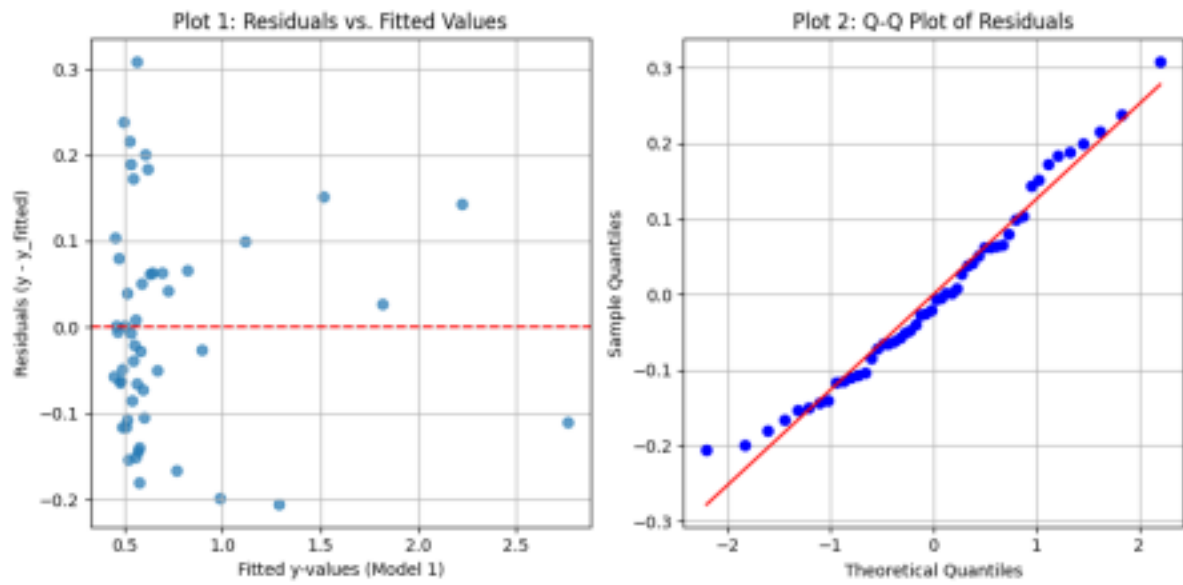


Figure 2: Diagnostic plots for Model 1: (Left) Residuals vs. Fitted Values, showing heteroscedasticity. (Right) Q-Q Plot, showing normality.

I used the program NLR project.py which I already mailed you.